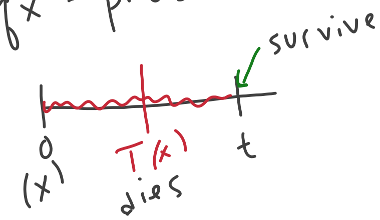
Chapter 8 Multiple State Models

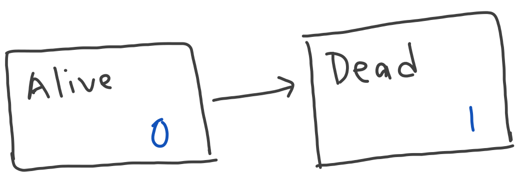
8.2.1 Alive-Dead Model

${}_tp_x$ = prob that (x) survives t years, or to age $x+t$

${}_tq_x$ = prob that (x) dies/fails during the next t years



$${}_tp_x + {}_tq_x = 1$$



Suppose we have a life age x $(x \geq 0)$ at $t=0$.

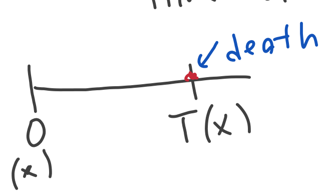
For each $t \geq 0$, we define the RV $Y(t)$ which here takes on the values 0 and 1.

$Y(t) = 0$ individual is alive at time t.

$Y(t) = 1$ individual is dead at time t, died before time t

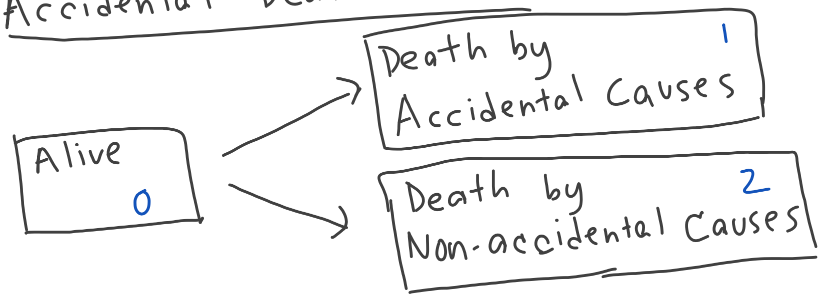
① The set of RVs $\{Y(t)\}_{t\geq 0}$ is a continuous time stochastic process.

② $T(x)$ = future lifetime RV
= time until death RV $\Big\}$ continuous RV



$$T(x) = \max\{t : Y(t) = 0\}$$

Accidental Death Model



- probabilities
- Insurance
- Annuities

Permanent Disability Model